NAME..... SCORING GUIDE 2026

PRE-UNEB EXAM

Uganda Advanced Certificate of Education

OCT/2026

S.6 BIOLOGY PRACTICAL

TIME: 3Hours:15minutes

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POST-MOCK SET 4 2026

BIOLOGY PAPER 2

INSTRUCTIONS TO CANDIDATES

Answers must be written in the spaces provided

Additional papers must not be inserted

TABLE OF SCORES

ITEMS	BASE CODES	SCORE LEVEL
1	DS	4
	DM	4
	DR	4
	ST	4
2	PE	4
	ME	4
	DP	4
	AN	4
	CR	4
	TOTAL	36

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ITEM 1

Jack is a s.5student at Jipra; who was entrusted with the authority to take care of the school rats in a cage that was found in the school laboratory; after about a year, he realized that one of the rats had escaped from the cage and has been wondering around the laboratory for about a year.

He finally captured the "stray rat" and returned it to the cage to live with other rats ("normal rats"). Jack fed the rats on pellets obtained from **specimen D**, which is believed to be rich with structures that when consumed by the rats provide **energy.**

Jack, monitored and observed the behavior of this stray rat while in the cage for some time and took records of the following observations;-

- The" stray rat" was able to climb and remain stable on high grounds unlike the rest of the rats. By doing so, it could gain access to extra food stored for the next meal.
- The "stray rat" was able to squeeze itself through tiny holes in the cage and get out searching for more food unlike the other rats.
- The stray rat ate larger amounts of food than the "normal rats"
- The stray rat was very difficult to catch because it ran much faster than the other rats.

In order for Jack to understand the differences in behavior and abilities of the stray rat and the other normal rat population, he made measurements of the **"normal rat"** population and recorded some data as shown in the table below;

Measurement	**length (mm)** Normal rat	Stray rat
Body without the tail	70	✓ 110mm
Tail	30	✓ 100mm
Circumference of the head at the widest part	50	✓ 75mm
Circumference of abdomen at widest part	60	✓ 120mm

He went ahead and dissected one rat from the normal population, he spread the Alimentary canal by placing the caecum to his far left and the ileum to his far right, he made some observations and measurements as shown in the table below.

DM-4

Structure	length in (mm)	
Ileum	200	960mm ✓
Blood vessels leaving the alimentary canal back to the liver	Fewer and short	Very many ✓

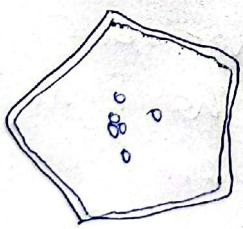
Tasks:

You have been provided with the stray **rat K**, **specimen D**, you are required to make similar observations and measurements on it as was done by Jack and use the data collected to explain why this rat is superior to the "normal rats".

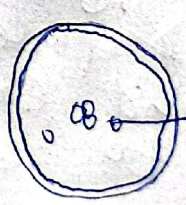
(a) Make a thin slide of **specimen D** and stain it with iodine solution, observe under medium power of the microscope.

Draw and label a single cell

A drawing of structures providing energy to ~~Normal~~ stray rat from specimen D viewed under medium power, ✓



or



X150, ✓

T - 1
M - 1
A - 1
L - 1

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(b) Make external measurements on the stray rat K as was done by jack on the normal rat, proceed to dissect the specimen to display structures similar to Jack's, make similar measurements and record your data.

Note: (Do not damage other structures in animal K you will use it in item2)

(i) Write your dissection procedures:

- Turned ~~the~~ animal K Ventral side upper most, and pinned in the feet.
- Lifted the abdominal skin and made an incision through the mid-ventral line and Continued upto the ~~diaphragm~~ lower jaw and towards the genitals.
- ~~M~~ Loosened the skin and pinned it aside to expose the body wall.
- Made another incision through the mid-ventral line of abdominal body wall and Continued upto the diaphragm and towards the genitalia.
- Pulled and the body wall aside and exposed the Viscera in the abdominal region.
- Displaced the liver lobes anteriorly, ileum and Caecum to the right of the animal to expose their Circulation.

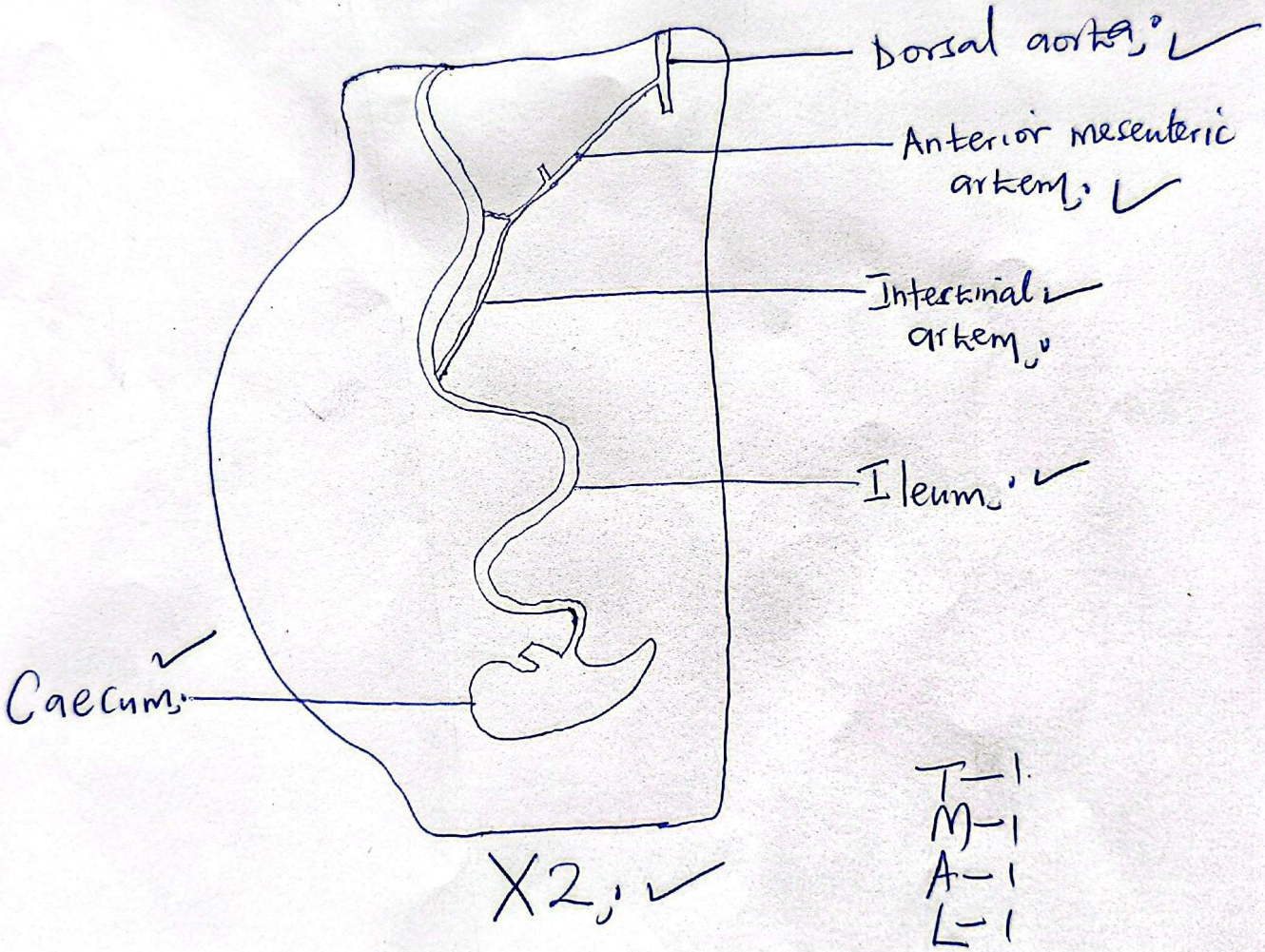
D5-4.

(ii) Determine the following ratios of measurements for both "normal rats" and "stray rats"

- Ratio of tail to body length
- Ratio of head circumference to abdominal circumference
- Ratio of ileum length to body length

(c) Make a drawing to show the internal structures that were spread and investigated by jack.

A drawing showing internal structures that were spread and investigated by jack of Animal K; ✓



T – 1
M – 1
A – 1
L – 1

(c) Using your data and observations collected; explain the suitability of the observations made about the "stray rat" by jack.

- Stray rats feed on pellets which contain food nutrients that provide it with energy ✓ used for making stray rats more capable of surviving ✓
- Stray rat has longer ✓ Ileum than normal rat ~~Not~~ increasing surface area for digestion ✓ and absorption of more food nutrients ✓ providing more energy to stray rat used for survival ✓
- ~~Animal K has larger ratio of body to~~
- Animal K has longer tail than normal rats, increasing surface area for maximum body balancing ✓ during locomotion hence stray rats being able to survive ✓
- Animal K has more blood circulation than normal rat increasing surface area for distribution of more important materials ✓ hence increasing chances of survival of stray rat K ✓

ST-4

ITEM 2

Dragon is a pharmaceutical industry manufacturing vitamin tablets for children, in order to make them sweet for easy chewing by children; they coat the tablet with sugars. These sugars are extracted from a **plant part D** using an enzyme extracted from the Alimentary canal of **animal K**. During the manufacture process, the mixture is kept at a **specific temperature** and a **specific ratio of water to the enzyme extract** must be used for effective production. Usually, food tests are done to determine the sugar levels and the green colour is most ideal for making the tablets.

The enzyme extract is made by crushing the animal pancreas in 20mls of water. The extract from **plant D** is made by crushing a 1cm by 1cm by 1cm cube of D in $20cm^3$ of water.

Musa is an intern who recently joined to train at the company. One day he was left alone to prepare the sugars. In the process he adjusted the temperature of the machines to $75°C$ and mixed the water and enzyme extract in the ratio of 5:1.

The tablets made did not meet the expected standard. Musa was concerned about this outcome and decided to carry out an investigation to discover the problem, he suspects the issue could have been with the temperature used and how he mixed the enzyme with water. He followed the following procedure in his investigation;

(i) Mixed water and enzyme in varying ratios as shown below, followed by $1cm^3$extract from **D** and kept the mixture at $40°C$ for 20 minutes.

Experiment	**1**	**2**	**3**	**4**	**5**
Volume of water (cm^3)	5	4	3	2	1
Volume of enzyme extract (cm^3)	1	2	3	4	5

(ii) He then chose the best outcome from the experiments above and used it to verify whether the temperature he used originally was also the problem, however due to limited time, he was unable to complete this part.

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Task:

You are required to carry out a similar investigation as done by Musa order to ascertain best ratio for the enzyme extract to water and whether the temperature initially used was not good.



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Task

D- starch from Irish extract
K- pancreas extract - Amylase

You have been provided with **animal K and plant part D** and other resources.
(Present your response in form a scientific report)

Aim:

An experiment to investigate the ascertain which ~~is the best ratio~~ of enzyme extract to water ~~mixed~~ in experiment 1 or 2 or 3 or 4 or 5; and the exact temperature suitable for the reaction in order to advise Musa.

Hypothesis:

Mixture ratio of experiment 1 heated at a temperature of 37°C - 40°C ~~we~~ lead to effective production of more sweets/sugars used for making sweet tablets. this is because in experiment 1 ~~th~~ the mixture ~~provid~~ contained the best enzyme concentration; and ~~tempe~~ temperature range 37°-40°C provided optimum temperature for the enzyme activity.

Variables:

Independent:- Concentration of enzyme in mixtures of experiment 1, 2, 3, 4 and 5.
- Temperature range.

Dependent:- Colour change.

Control:- Temperature, time.
- Volume of enzyme extract.
- Volume of water.
- Volume of food nutrient D.
- Volume of test reagents.

PE-4

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Risks and Mitigation

– Burns from hot test tube mitigated by using test tube holders; ✓

– Cuts from broken test tubes mitigated by careful handling; ✓

Procedures

1. Labelled test tubes 1 2 3 4 5 ✓R
2. To test tube 1 added the mixture as shown the table able ✓R
3. Repeated the above step 2 for ~~procedur~~ experiments 2, 3, 4 and 5 ✓A
4. To each mixture was added 2 cm^3 of extract D in the test tube ✓R
5. Each mixture in the test tube was incubated ✓R under water bath maintained under temperature range of 37°C–40°C ✓A for 30 minutes. ✓A
6. After incubating, each mixture was tested for ~~Ben~~ reducing sugars by adding 2 cm^3 of Benedict's Solution ✓R and boiled; ✓R

ME – 4

Data presentation

Experiment	observation
1	Turbid solution turned to pale blue solution; ✓
2	Turbid solution turned to pale blue/ green solution; ✓
3	Turbid solution turned to pale blue to green solution; ✓
4	Turbid solution turned to pale blue, green solution to yellow ppt; ✓
5	Turbid solution turned to pale blue, green solution to yellow ppt to ~~org~~ orange ppt; ✓

DP-4

Analysis and explanation:

Experiment 3 provided appropriate concentration of enzyme to break down starch to little reducing ~~sg~~ sugars; ✓
Temperature range of 37°C - 40°C provided optimum temperature for proper functioning of active substance; ✓ which provided kinetic energy that increased the rate of collision between the active substance and starch; ✓ leading to formation of little reducing sugars; ✓
High temperature of 75°C, denatured ✓ the active ~~act~~ substance from extract of pancreas of animal K hence no formation of reducing sugars from ✓ Catalytic break down of starch.

AN - 4

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Conclusion:

Mixture of $3cm^3$ of water and $3cm^3$ of enzyme extract ~~with~~ ~~treated~~ reacted with food nutrients in extract D at optimum temperature 37°-40°C provides the best sweets for formation of ~~tent~~ tablets.

Recommendation:

I would advise Musa to apply a mixture of $3cm^3$ of water and $3cm^3$ of enzyme with extract D to generate best sweets for tablets. ~~at~~ incubated at a temperature of 37°C - 40°C.

CR-4

END